

Biogas Conditioning Skid - Process Design

LETEK DCG / Bachoco layer farms, Merida, Yucatan - chicken manure digestate dryer, 140 MT/day case

Midwest Custom Engineering, Inc.
Doc. MCE-LETEK-BGS-001 Rev A - 12 Sep 2026
Prepared by: J. Shipley - Budgetary design

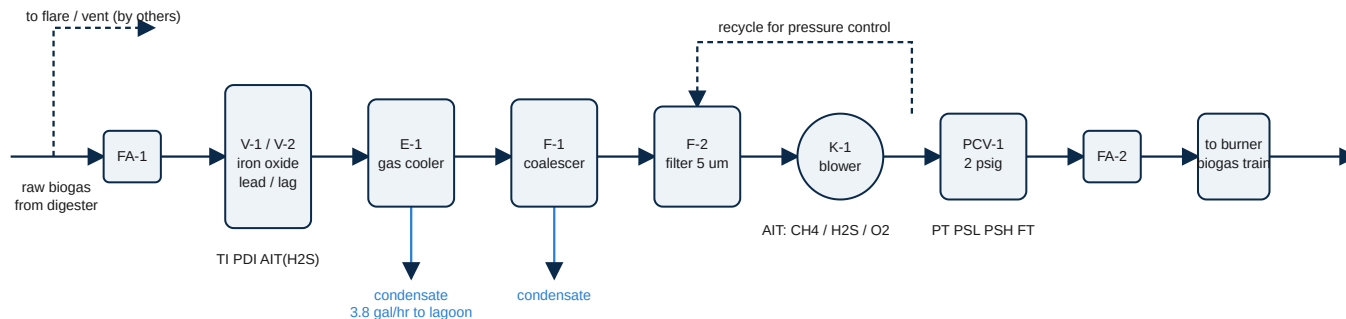
1. Purpose and design basis

Raw digester biogas is saturated with water, carries hydrogen sulfide, and arrives at a few inches of water column. The dryer burner needs dry, low-sulfur gas at a steady 2 psig. The skid does three things in order: bulk H₂S removal, moisture removal, and pressure boosting, with flame arresters at both ends and a control panel that hands the burner management system a clean permissive.

Item	Value	Basis
Biogas flow, design case (140 MT/day)	7,455 Nm ³ /day = 193 scfm	Customer figure, 19 Aug 2026
Skid design flow	232 scfm (+20%)	160 MT/day case and digester swings
Composition	55% CH ₄ , 45% CO ₂ , saturated	Customer figure; analysis pending
Heating value at 193 scfm	5.8 MMBtu/hr LHV (139 MMBtu/day), 6.5 HHV	911 / 1,012 Btu/scf CH ₄
Share of dryer duty	80% of 7.3 MMBtu/hr normal fire; 58% of 10 MMBtu/hr max	LHV basis; LPG makes up the balance
H ₂ S inlet (assumed)	up to 5,000 ppmv = 5.2 lb/hr H ₂ S, 117 lb/day S	Poultry manure digestate is sulfur-rich; analysis pending
H ₂ S outlet	< 200 ppmv	Burner train, condensate and stack corrosion
Inlet temperature / pressure	95-100 F, 2-8 in WC	Mesophilic digester; to be confirmed
Outlet to burner	2.0 psig, dew point ≥ 20 F below ambient, < 5 micron particulate	NFPA 86 train, no condensate in the valve train
Site	30 ft elevation, 85-100 F, coastal, wet season	Outdoor skid under a roof

2. Process flow

Stream: 232 scfm design (193 normal) - 100 F sat. - 5,000 ppm H₂S -> 40 F dew point, < 200 ppm H₂S, 2 psig



3. Sizing

Tag	Equipment	Sizing	Result / selection
FA-1, FA-2	In-line deflagration flame arresters, 316 SS elements	232 scfm, 4 in line, IIA gas group	4 in bidirectional, inlet and outlet of skid
V-1, V-2	Iron-oxide H ₂ S vessels, lead/lag, 304 SS, top-loading, wet change-out	60 s empty-bed contact at design flow: 232 ft ³ per vessel; superficial velocity 12 fpm	2 x 5 ft dia x 14 ft T/T, 12 ft bed, about 14,000 lb pelletized iron oxide each (0.15 lb S per lb media)
Media life, lead vessel		117 lb S/day at 5,000 ppm = 780 lb media/day	18 days at 5,000 ppm; 179 days at 500 ppm. See note below.
E-1	Gas cooler, shell and tube, 316L tubes, glycol chilled	874 lb/hr gas, 100 F to 40 F: 15,800 Btu/hr sensible + 33,400 latent = 49,300 Btu/hr	5-ton air-cooled glycol chiller; removes 32 lb/hr water (3.8 gal/hr); outlet water 4 lb/hr
F-1	Coalescing separator with level-controlled condensate trap, 316L	Liquid-out at design flow, 0.5 in WC drop	4 in coalescer, automatic drainer to condensate line
F-2	Particulate filter, 5 micron, with PDI	232 scfm	Duplex cartridge housing so a change-out does not stop the dryer

K-1	Rotary-lobe biogas blower, gas-tight, explosion-proof motor, VFD	232 scfm from 0 to 2.5 psig; 4.6 shaft hp at 55%	7.5 hp, 480/3/60, VFD on outlet pressure; 20 F of blower heat gives the dew-point margin after cooling
PCV-1	Outlet pressure control valve and recycle valve to blower suction	2.0 psig set, 10:1 turndown with the burner	2 in PCV, 2 in recycle, PT/PSL/PSH to the BMS
AIT-1/2	Gas analyzers, inlet and outlet	CH ₄ , H ₂ S, O ₂	Outlet H ₂ S high alarm at 200 ppm, trip to LPG at 500 ppm; O ₂ > 2% trip
LCP-1	Local control panel, NEMA 4X 304 SS	VFD, analyzer inputs, valve outputs, Modbus to burner BMS and plant PLC	Permissives to the burner: gas pressure OK, H ₂ S OK, O ₂ OK, arrester temperature OK
Skid	Structural steel skid, 304 SS piping (wet side 316L), 4 in main line	Roughly 10 ft x 24 ft footprint, chiller alongside	Ship as two modules: vessels; cooler/blower/controls

H₂S loading drives the design. At 5,000 ppm the dry media alone would consume about 780 lb per day, which is neither economical nor practical to change out. The recommended arrangement is bulk removal in the digester by micro-aeration (2 to 4% air into the headspace) or ferric chloride dosing, both by the digester supplier, bringing the gas to 500 ppm or less. The iron-oxide vessels then run as a polishing stage with a 6-month media life. If the digester cannot accept micro-aeration, a biological trickling scrubber ahead of the vessels is the alternative and roughly doubles the skid footprint. The biogas analysis decides this; the vessels and cooler stay the same either way.

4. Mass and energy summary (193 scfm normal flow)

Stream	Flow	T	P	Condition	Utility	Demand
1 Raw biogas	193 scfm, 874 lb/hr gas + 36 lb/hr water	100 F	4 in WC	saturated, 5,000 ppm H ₂ S	Electrical, blower K-1	7.5 hp, 5.6 kW at design
2 After V-1/V-2	192 scfm	100 F	2 in WC	< 200 ppm H ₂ S, saturated	Electrical, chiller	5 ton, about 6 kW
3 After E-1 / F-1	192 scfm gas, 4 lb/hr water	40 F	0 in WC	40 F dew point	Electrical, controls and analyzers	1 kW
4 After K-1 / PCV-1	192 scfm	60-70 F	2.0 psig	dew point 20-30 F below gas temperature	Total connected	approx. 13 kW, 480/3/60
C Condensate	32 lb/hr, 3.8 gal/hr	40 F	-	acidic, to lagoon or digester	Nitrogen for purge and change-out	bottled, 2 cylinders on site
S Spent media	14,000 lb per change-out	-	-	non-hazardous after air passivation; landfill	Instrument air	none (electric actuators)
					Condensate disposal	3.8 gal/hr, gravity to lagoon

5. Safety and controls

- Flame arresters at skid inlet and outlet, temperature switches on both; the outlet arrester is the last device before the burner biogas train.
- Classified area: Class I Division 2 around the skid; blower motor and instruments rated accordingly; skid under a roof, open sides, no enclosure.
- Trips to LPG (automatic changeover in the burner BMS): low gas pressure, high H₂S, high O₂, high blower discharge temperature, arrester high temperature, chiller failure with dew point alarm.
- Iron-oxide media is pyrophoric when spent and dry; vessels are changed out wet with nitrogen purge, one vessel at a time (lead/lag lets the dryer keep running).
- Condensate is acidic (H₂S, CO₂); 316L on the cooler, coalescer and condensate piping, 304 SS elsewhere on the skid.
- Pressure relief on the blower discharge to the flare/vent header; the digester's own flare handles any gas the dryer cannot take.

6. Information needed to finalize

Item	Why it matters	From
Biogas analysis: CH ₄ , CO ₂ , H ₂ S, O ₂ , N ₂ , moisture, siloxanes	Sets the H ₂ S removal stage (in-digester vs. skid), media life, and whether an activated-carbon polish is needed	LETEK / digester supplier
Digester gas pressure and holder volume	Blower suction design, whether the holder can buffer burner turndown	Digester supplier
Can the digester accept micro-aeration or FeCl ₃ dosing?	Decides whether the skid is a polish stage (as sized) or needs a biological scrubber	Digester supplier
LPG supply pressure and tank location	Burner changeover and train design	LETEK
Site plan: digester to dryer distance	Line size and blower head; 4 in line assumed for up to 300 ft	LETEK

7. Assumptions and confidence

Confidence is **medium** for the cooler, blower and controls (these follow from the flow and composition the customer gave) and **low** for the H₂S stage until the analysis arrives, because a 5,000 ppm assumption versus a 1,500 ppm reality changes the media economics by a factor of three and decides whether in-digester treatment is required. Pricing for this skid follows component quotations and is not included in this document.